Honors Calculus 1610

Review for Exam 2

October 30, 2026

Which practice problems felt hardest?

We sorted the Exam 2 practice problems by how they felt.

1. High ("no idea where to start")

Problems 5, 14, 7.

2. Medium ("I think maybe I could?")

Problems 13, 11, 12, 10, 3.

3. "OK, I guess"

The rest.

We went through 5, 7, 14, 12 and 10.

Problem 5: negating uniform continuity

Uniformly continuous

$\forall \epsilon > 0 \;\; \exists \delta > 0$ such that $\forall x_1, x_2$: $\quad |x_1 - x_2| < \delta \implies |f(x_1) - f(x_2)| < \epsilon.$

The shape is "for all $\epsilon > 0$, something is true."
So the negation is "for some $\epsilon > 0$, that thing is false."

Not uniformly continuous

$\exists \epsilon > 0 \;\; \forall \delta > 0 \;\; \exists x_1, x_2$ such that $|x_1 - x_2| < \delta$ but $|f(x_1) - f(x_2)| \geq \epsilon.$

For $f(x) = 1/x$ on $(0, 1)$: try $x_1 = \frac{1}{n}$ and $x_2 = \frac{1}{2n}$.

Problem 7: the symmetric difference quotient

Add and subtract $f(c)$:

$$\lim_{h\to 0}\frac{f(c+h)-f(c-h)}{2h}=\lim_{h\to 0}\frac{f(c+h)-f(c)+f(c)-f(c-h)}{2h}.$$

The first half is the derivative: $\lim_{h\to 0}\dfrac{f(c+h)-f(c)}{h}=f'(c)$.

For the second half, substitute $k=-h$; as $h\to 0$, also $k\to 0$:

$$\lim_{h\to 0}\frac{f(c-h)-f(c)}{h}=-\lim_{k\to 0}\frac{f(c+k)-f(c)}{k}=-f'(c).$$

So the whole thing is $\frac{1}{2}f'(c)+\frac{1}{2}f'(c)=f'(c)$.

Problem 14(a): $\lim_{t\to 0}(t, t^2) = (0, 0)$

We need $\|(t, t^2) - (0, 0)\| = \sqrt{t^2 + t^4} = |t|\sqrt{t^2 + 1} < \epsilon$.

Both sides are nonnegative, so we can square:

$$|t|^2(t^2 + 1) < \epsilon^2 \quad \Longleftrightarrow \quad |t|^4 + |t|^2 < \epsilon^2.$$

It is enough to make each term less than $\epsilon^2/2$:

$$|t|^4 < \frac{\epsilon^2}{2} \text{ when } |t| < \sqrt[4]{\frac{\epsilon^2}{2}} = \frac{\sqrt{\epsilon}}{\sqrt[4]{2}}, \qquad |t|^2 < \frac{\epsilon^2}{2} \text{ when } |t| < \sqrt{\frac{\epsilon^2}{2}} = \frac{\epsilon}{\sqrt{2}}.$$

So $\delta = \min\left\{\sqrt{\epsilon}/\sqrt[4]{2},\ \epsilon/\sqrt{2}\right\}$ works.

Problem 12(a): the evaluation rule

Evaluation rule

If F is an antiderivative of f (and f is continuous), then

$$F(b) = \int_a^b f(t)\,dt + F(a), \qquad \text{or} \qquad F(b) - F(a) = \int_a^b f(t)\,dt.$$

Write $\frac{2}{x^2} = 2x^{-2}$, and run the power rule backwards:

$$\int_1^2 \Big(3x^2 - \frac{2}{x^2}\Big)dx = \Big[\underbrace{\tfrac{3}{3}x^3 + 2x^{-1}}_{\text{antiderivative}}\Big]_1^2 = (8+1) - (1+2) = 6.$$

Problem 10(b): the setup

f is differentiable on (a, b), f' is differentiable at $x = c$,

$$f'(c) = 0, \qquad f''(c) > 0.$$

Want: for some $h > 0$, $f'(x) > 0$ on $(c, c + h)$.

Since $f'(c) = 0$, the definition of $f''(c)$ reads

$$\lim_{k \to 0} \frac{f'(c + k) - f'(c)}{k} = \lim_{k \to 0} \frac{f'(c + k)}{k} = f''(c) > 0.$$

Problem 10(b): the argument

Suppose no such h exists. That means: for every $h > 0$ there is some $x \in (c, c+h)$ where $f'(x) \le 0$.

But take $\epsilon = f''(c) > 0$ in the limit above: there is a $\delta > 0$ with

$$0 < |k| < \delta \implies \left| \frac{f'(c+k)}{k} - f''(c) \right| < f''(c) \implies \frac{f'(c+k)}{k} > 0.$$

Now take $h = \delta$. Any $x \in (c, c+\delta)$ is $x = c + k$ with $0 < k < \delta$, so $f'(x)/k > 0$ and $k > 0$ give $f'(x) > 0$ — contradicting the supposition. $\square$

The same argument with $-\delta < k < 0$ gives $f'(x) < 0$ on $(c-\delta, c)$.